Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

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Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

Autonomous vehicles depend extensively on large-scale annotated datasets to train machine learning models for tasks such as object detection, environmental perception and drivingrelated decision-making. Traditionally, such datasets are gathered from real-world driving environments using sensors including cameras, LiDAR and radar. However, real-world datasets frequently contain sensitive information such as pedestrian identities, vehicle registration details and geographic location data, thereby creating substantial privacy and ethical concerns when such data is shared or published for research and development purposes.

Although simulation platforms such as CARLA and AirSim enable researchers to generate synthetic datasets for autonomous vehicle training, most existing approaches primarily emphasize simulation realism and dataset diversity while giving comparatively limited consideration to privacy preservation or pipeline security. Recent research has further highlighted the growing cybersecurity risks associated with autonomous vehicle data collection, storage and sharing, including threats such as data tampering, unauthorized access and re-identification attacks [18], [20].

This research proposes a privacy-enhanced and cybersecurity-hardened synthetic data generation framework for autonomous vehicle training using the CARLA simulator and Waymo Open Dataset. The proposed system is designed to generate realistic driving scenarios and capture sensor data, including RGB images, depth maps and segmentation outputs. Privacy-preserving mechanisms such as identity abstraction and controlled noise injection will be incorporated during the data generation process. Additionally, a dedicated cybersecurity layer — encompassing SHA-256 data integrity verification, AES-256 encryption, and rolebased access control — will be integrated directly into the data generation pipeline to protect against unauthorized access and tampering.

The anticipated outcome is a framework that facilitates safer data-sharing practices and contributes to the development of reliable machine learning models for autonomous driving systems, while addressing both privacy and data security requirements identified in recent literature [17], [19], [21].

Keywords: Autonomous Vehicles, Synthetic Data Generation, Privacy Preservation, CARLA Simulator, Cybersecurity, Data Integrity, Differential Privacy, Machine Learning

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List of Abbreviations

AES	Advanced Encryption Standard
AI	Artificial Intelligence
AirSim	Aerial Informatics and Robotics Simulation
API	Application Programming Interface
AV	Autonomous Vehicle
CARLA	Car Learning to Act
CV	Computer Vision
DP	Differential Privacy
DSR	Design Science Research
FR	Functional Requirement
GDPR	General Data Protection Regulation
HMAC	Hash-based Message Authentication Code
LiDAR	Light Detection and Ranging
ML	Machine Learning
OASim	Open and Adaptive Simulator
RBAC	Role-Based Access Control
RGB	Red, Green, Blue
SDG	Sustainable Development Goal
SHA	Secure Hash Algorithm
TLS	Transport Layer Security
WBS	Work Breakdown Structure

1. Introduction

1.1 Background and Context

Autonomous vehicles (AVs) depend on substantial volumes of training data to develop perception and decision-making models for tasks including object detection, lane detection and traffic sign recognition. Conventionally, these datasets are obtained from real-world driving environments through sensors such as cameras, LiDAR and radar. Nevertheless, the collection of real-world datasets is costly, time-intensive and associated with significant privacy and security concerns, as the data may include identifiable pedestrians, vehicle registration plates, and location-related information [18].

To address these challenges, simulation platforms have emerged as an essential instrument in autonomous driving research. One of the most extensively utilized simulators is CARLA, which provides realistic urban driving environments, dynamic weather conditions, and multiple sensor outputs for autonomous vehicle testing and data acquisition [1]. CARLA enables researchers to generate labeled datasets and assess perception algorithms under controlled conditions. Recent studies have further examined the advantages and implementation challenges associated with CARLA in autonomous driving research, emphasizing its capacity to generate large-scale synthetic datasets while supporting diverse traffic scenarios [3], [4].

The Waymo Open Dataset, created by Waymo, is another widely adopted tool in autonomous driving research. It includes large-scale real-world driving data gathered with sensors such as LiDAR, cameras and radar, and has been continuously expanded with new labels and benchmarks through to 2024 [14]. Researchers combine simulation platforms such as CARLA with real-world datasets such as Waymo to improve dataset diversity and training realism [15].

Recent studies have demonstrated that synthetic datasets can substantially improve the performance of deep learning models for autonomous vehicles, reducing the domain gap between simulated and real-world environments and enabling the representation of rare or hazardous driving scenarios [6], [16]. The SHIFT dataset, released in 2022 and generated using CARLA, is among the largest multi-task synthetic datasets for autonomous driving, demonstrating the growing maturity of simulation-based data generation [15]. Survey studies have further highlighted persistent challenges in synthetic data research, including realism, domain adaptation, and dataset quality [7], [8].

In parallel, the security risks associated with autonomous vehicle data pipelines have received increasing attention. Gao et al. [17] conducted a comprehensive review of autonomous driving security, identifying data tampering, sensor spoofing and unauthorized access as critical threats to AV systems. Similarly, Liao et al. [18] examined the multifaceted data security challenges facing autonomous driving, noting that data leakage, storage vulnerabilities, and reidentification risks remain largely unaddressed in current pipelines.

Privacy-preserving machine learning techniques, including differential privacy, have been proposed to safeguard sensitive information during model training [9], [10]. Federated learning combined with differential privacy has been applied to autonomous driving collaborative training to protect sensitive data [11], [19]. A comprehensive survey on privacy in autonomous vehicles identified that privacy-aware data generation and pipeline security remain underexplored areas requiring dedicated research [20].

Overall, the existing body of research demonstrates the considerable potential of simulation platforms and synthetic datasets for training autonomous vehicle models. However, the majority of current approaches primarily emphasize simulation realism and model performance, while privacy protection and data pipeline security are frequently treated as secondary concerns.

1.2 Literature Review

Autonomous vehicles depend on large-scale datasets to train machine learning models for perception-related tasks, including object detection, lane detection and environmental understanding. Benchmark datasets such as KITTI and Cityscapes have made significant contributions to the field by providing labeled urban scene data for the training and evaluation of computer vision models [4], [5]. Nevertheless, the collection of real-world datasets is costly, time-consuming, and raises serious privacy and security concerns, as recorded data may contain identifiable pedestrians, vehicles and geographic information [18].

Simulation platforms have been introduced as an effective alternative for generating synthetic datasets. CARLA provides realistic urban driving environments with dynamic traffic conditions, varying weather patterns and multiple sensor outputs [1]. AirSim provides a highfidelity simulation environment built on Unreal Engine, supporting both autonomous vehicles and drones [2]. These platforms allow researchers to recreate hazardous or rare driving scenarios without real-world risks, and recent work has demonstrated their use in generating large, diverse datasets such as SHIFT — one of the largest multi-task synthetic datasets for autonomous driving, produced entirely within the CARLA simulator [15].

Domain adaptation research has further addressed the gap between synthetic and real-world data. Mullick et al. [16] proposed novel approaches for adapting synthetic driving datasets to real-world conditions using GAN-based domain translation, demonstrating the practical value of synthetic data when properly adapted. Survey studies have emphasized the increasing significance of synthetic datasets for autonomous driving and identified ongoing challenges in realism and domain gaps [7], [8].

Privacy-preserving machine learning techniques, including differential privacy, have been proposed to protect sensitive information during training and data analysis [9], [10]. Emerging research has applied these techniques directly to autonomous driving; Han et al. [19] proposed a personalized federated learning scheme with correlated differential privacy for autonomous driving, demonstrating that privacy guarantees can be maintained without significant loss of model utility. A comprehensive survey on privacy in autonomous vehicles [20] reviewed the full spectrum of protection methods, from anonymization and differentially private learning to cryptographic approaches. Recent reviews have also examined the broader security and privacy challenges facing autonomous driving systems, identifying attack vectors and defense mechanisms across perception, communication, and data pipeline stages [17], [21].

1.3 Research Gap

Despite the substantial progress achieved in simulation-based autonomous vehicle research, several critical limitations persist. Simulation platforms such as CARLA and the Waymo Open Dataset have effectively facilitated large-scale synthetic dataset generation [1], [14]. However,

the majority of existing simulation frameworks primarily focus on environmental realism and sensor fidelity rather than incorporating privacy protection or security controls during the dataset generation process itself.

Recent surveys on synthetic datasets for autonomous driving have predominantly examined model performance and domain generalization [7], [8], [16]. Although these studies demonstrate the value of synthetic data, they seldom address the security risks of the data generation pipeline. As autonomous vehicle research grows more collaborative, datasets are frequently shared among institutions and industry partners, intensifying concerns around data integrity, unauthorized access and re-identification attacks [18], [20].

Liao et al. [18] and Gao et al. [17] both identify that data security in autonomous driving pipelines remains inadequately addressed, with most protection mechanisms applied only postgeneration. Common post-processing techniques such as anonymization and image blurring do not provide strong security guarantees and fail to embed protection directly into the generation stage. The comprehensive security review by Giannaros et al. [21] further highlights that existing frameworks lack integrated pipeline-level security controls.

Although privacy-preserving techniques such as differential privacy and federated learning have been studied extensively [9], [10], [19], their integration with secure synthetic dataset generation remains limited. Han et al. [19] addressed privacy in collaborative AV training, and the survey by Petit and Shladover [20] reviewed privacy methods for AV data; however, neither addresses cybersecurity controls at the point of simulation-based dataset generation and storage.

Therefore, a clear research gap exists in the development of a framework that integrates cybersecurity controls — including data integrity verification, encryption, and access control — directly into the simulation-based data generation pipeline. This research seeks to address this limitation by developing a secure synthetic data generation component within the CARLAbased framework, where SHA-256 hashing, AES-256 encryption and role-based access control are embedded during generation to improve both dataset security and usability.

2. Objectives

2.1 Main Objective

To design and implement a privacy-enhanced and cybersecurity-hardened synthetic data generation framework using the CARLA simulator and Waymo Open Dataset for autonomous vehicle training, thereby enabling the creation of realistic, privacy-preserving, and secure datasets for machine learning model development.

2.2 Specific Objectives

- Configure the CARLA simulation environment to generate realistic autonomous driving scenarios.
- Utilize Waymo Open Dataset to support dataset diversity and validation of generated synthetic data.
- Integrate privacy-preserving mechanisms during the synthetic data generation process to reduce the risk of sensitive information exposure.
- Implement a cybersecurity layer within the data generation pipeline, including SHA256 data integrity verification, AES-256 encryption of stored datasets, and role-based access control (RBAC) on simulation outputs.
- Evaluate the usefulness of the generated synthetic datasets for training machine learning models in autonomous vehicle perception tasks.
- Analyze the trade-off between dataset realism, privacy protection, security overhead and model performance in autonomous vehicle training.

3. Methodology

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach, which emphasizes the design and evaluation of an artefact intended to address a real-world problem. In the present study, the artefact is a privacy-enhanced and security-hardened synthetic data generation framework for autonomous vehicle training. The DSR approach is appropriate because the study seeks to develop a system that integrates simulation-based data generation with both privacy-preserving and cybersecurity mechanisms, addressing the pipeline-level security gaps identified in recent literature [17], [18].

The research process comprises several stages: problem identification, system design, implementation of the simulation framework, dataset generation, security integration, and evaluation. The proposed framework will be implemented using the CARLA simulator and Waymo Open Dataset to produce realistic autonomous driving scenarios while incorporating privacy-preserving and security mechanisms throughout the data generation process.

3.2 Technical Methods and Algorithms

The proposed system will generate synthetic datasets for autonomous vehicle perception tasks through a simulation-based pipeline. The CARLA Python API will be utilized to capture sensor outputs — including RGB images, depth maps and segmentation data — and organize them into labeled dataset batches automatically.

Privacy-preserving techniques including identity abstraction, texture modification and controlled noise injection will be applied to prevent identifiable information from being embedded within generated datasets, consistent with approaches surveyed in recent privacy literature [20].

A dedicated cybersecurity layer will be implemented as part of the data generation component. SHA-256 cryptographic hashing will be applied to each dataset batch immediately upon capture to verify data integrity and detect tampering — a critical control given the data integrity threats identified in autonomous driving security reviews [17]. AES-256 encryption will protect stored datasets at rest and during inter-component transfer. Role-based access control (RBAC) will restrict access to raw simulation outputs to authorized pipeline modules and personnel only, reducing the attack surface of the data generation pipeline [18], [21].

3.3 Tools and Platforms with Justifications

The CARLA Simulator will serve as the main simulation platform for generating autonomous driving environments. CARLA offers diverse sensor outputs, dynamic traffic scenarios and realistic urban surroundings, and has been used in generating large-scale synthetic datasets including SHIFT [15].

The Waymo Open Dataset will provide a real-world reference for improving dataset diversity and validation, offering high-quality LiDAR and camera data continuously updated through recent releases [14].

The Python programming language will be used to access the CARLA API and automate dataset generation. Python cryptographic libraries — including hashlib for SHA-256 and

PyCryptodome for AES-256 — will implement the security layer. These are well-established, open-source tools widely used in secure data pipeline design.

3.4 Data Collection and Ethical Considerations

All datasets will be generated entirely within the CARLA simulation environment, eliminating the need to collect real-world data containing sensitive personal information. This approach supports compliance with data privacy regulations including GDPR, as recommended in recent autonomous driving data security research [18]. Privacy-preserving and cryptographic security mechanisms will be integrated into the generation process to ensure responsible data management throughout the pipeline.

3.5 Validation Strategy and Performance Metrics

The generated synthetic datasets will be evaluated for realism, diversity and machine learning utility. Models trained on the synthetic data will be assessed on perception tasks including object detection and scene segmentation, using metrics such as accuracy, precision, recall, mean Average Precision (mAP) and Frechet Inception Distance (FID). The impact of privacypreserving mechanisms and security controls on dataset usability will be examined, consistent with the privacy-utility trade-off analysis approach applied in recent federated learning research for autonomous driving [19]. Security validation will include hash verification tests on generated batches and RBAC access control audits.

4. High-Level System Architecture

4.1 Overview of the Proposed System

The proposed system generates privacy-enhanced and cybersecurity-hardened synthetic datasets for autonomous vehicle training using the CARLA simulation platform combined with the Waymo Open Dataset for validation. It incorporates a simulation environment, a scenario generation module, a secure synthetic data generation component, a privacy protection module and a dataset storage framework. These components collectively produce realistic driving scenarios while ensuring that privacy-preserving mechanisms and cybersecurity controls — addressing threats identified in recent security literature [17], [21] — are applied throughout the data generation process.

4.2 System Components

The proposed architecture consists of several principal components.

CARLA Simulation Environment

The CARLA simulator functions as the core simulation platform responsible for generating virtual driving environments, creating realistic road conditions, traffic scenarios, pedestrians, and environmental variations such as weather and lighting [1].

Scenario Generation Module

This module creates diverse driving scenarios incorporating variations in traffic density, road types and environmental conditions, ensuring the generated dataset contains representative driving situations.

4.2.1 Secure Synthetic Data Generation Component (IT22541284)

This component constitutes the primary research focus of IT22541284. It captures sensor data from the CARLA simulation environment, converts it into labeled datasets for machine learning model training, and secures the entire data generation pipeline through an integrated cybersecurity layer — directly addressing the pipeline-level security gaps identified in recent autonomous driving security research [17], [18], [21].

The component interacts with the CARLA Python API to automatically record RGB images, depth maps and segmentation data and organize them into structured, labeled dataset batches. Three cybersecurity controls are embedded within the pipeline:

i. Data Integrity Verification

Each dataset batch is subject to SHA-256 cryptographic hashing immediately upon capture. A hash manifest file is created alongside each batch, and hash values are re-verified before any downstream processing. Mismatches flag potential tampering or corruption for quarantine and review. This mechanism provides end-to-end integrity assurance, addressing the data tampering threats documented in autonomous driving security literature [17].

ii. AES-256 Encryption of Dataset Batches

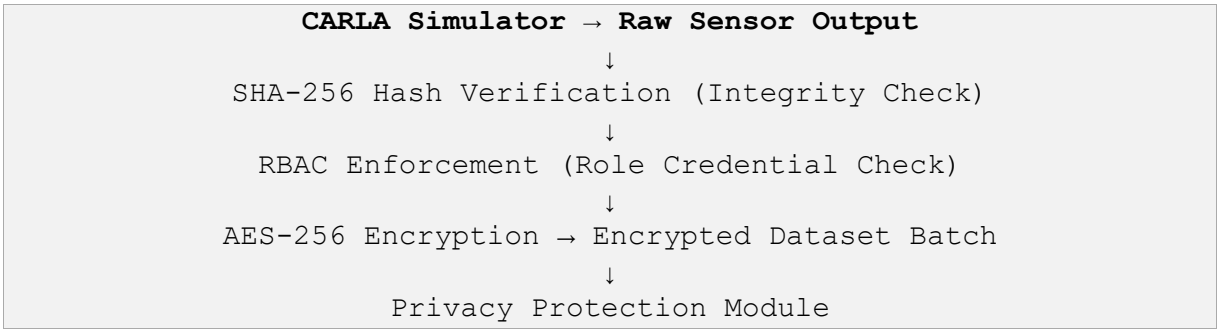
Following integrity verification, all dataset batches are encrypted using AES-256 in CBC mode prior to storage or inter-component transfer. Key management is handled separately from

storage to prevent unauthorized access even if the storage medium is compromised. This control aligns with data protection best practices and supports compliance with GDPR and related regulations [18].

iii. Role-Based Access Control (RBAC) on Simulation Outputs

Three roles are defined within the pipeline: Simulation Engineer (read/write access to simulation outputs), Privacy Engineer (read access to encrypted batches for privacy processing), and ML Researcher (read access to processed datasets only). Access control is enforced programmatically within the CARLA API wrapper. This limits the attack surface of the pipeline and prevents unauthorized data extraction, consistent with security controls recommended in recent comprehensive AV security reviews [21].

Figure 2: Secure Synthetic Data Generation Pipeline (IT22541284)



Privacy Protection Module

Applies identity abstraction, object anonymization and controlled noise injection to the encrypted batches to prevent privacy leakage while preserving dataset usability, consistent with privacy-preserving approaches surveyed in recent AV privacy literature [20].

Dataset Storage and Management

Stores datasets in structured formats for machine learning training, managing metadata including scenario parameters, simulation configurations and hash manifest files for integrity auditing.

4.3 Component Interaction

The components interact through a sequential, security-aware data generation pipeline. The scenario generation module configures the simulation environment within CARLA. CARLA produces realistic driving scenes and sensor outputs, which are captured by the Secure Synthetic Data Generation Component. Integrity verification, RBAC checks and encryption are applied before the secured data is forwarded to the privacy protection module. Privacyprocessed datasets are then stored for machine learning model training.

4.4 Data Flow Explanation

The system follows a structured data flow process:

- The scenario generator creates driving scenarios within the CARLA simulation environment.

- CARLA generates virtual environments and produces raw sensor outputs.
- The secure synthetic data generator captures images and sensor data from the simulation.
- SHA-256 hashing is applied to each dataset batch to verify integrity.
- RBAC credentials are verified before any data read or write operation proceeds.
- AES-256 encryption is applied to the verified dataset batch.
- The encrypted batch is forwarded to the privacy protection module.
- Processed datasets are stored in the dataset storage system.
- Datasets are subsequently used for machine learning model training and evaluation.

4.5 Scalability and Security Considerations

The system is designed with scalability in mind, supporting large-scale dataset generation through multiple parallel simulation scenarios. The modular architecture supports straightforward integration of additional simulation environments, sensors or privacy mechanisms in future extensions.

From a security perspective, the cybersecurity controls — SHA-256 integrity verification, AES-256 encryption and RBAC — are embedded directly into the pipeline, addressing the data security threats and vulnerabilities identified in recent comprehensive reviews of autonomous driving security [17], [21] and data security research [18]. By generating synthetic data within a simulation environment and applying both privacy-preserving and cryptographic controls throughout generation, the system minimizes risks associated with real-world data collection and collaborative dataset sharing.

5. User Requirements

5.1 Stakeholder Identification

The proposed system involves several stakeholders who interact with or benefit from the framework.

Autonomous Vehicle Researchers

Researchers utilize the generated synthetic datasets to train and evaluate machine learning models for perception and decision-making tasks.

Artificial Intelligence Developers

AI developers use the datasets to design and assess deep learning models for autonomous driving systems.

Simulation Engineers

Simulation engineers configure and manage the simulation environment to generate realistic driving scenarios.

Cybersecurity Engineers

Cybersecurity engineers configure, monitor and audit the security controls embedded within the data generation pipeline, including integrity verification, encryption key management and RBAC policies — roles increasingly recognized as essential in AV data security frameworks [18], [21].

Technology Companies and the Automotive Industry

Organizations engaged in autonomous vehicle development may employ synthetic datasets to improve their machine learning models without relying on real-world data collection.

5.2 Functional Requirements

The system shall provide the following functional capabilities:

- FR1: The system shall generate realistic autonomous driving environments using the CARLA simulator.
- FR2: The system shall create multiple driving scenarios incorporating variations in traffic conditions, road structures and environmental settings.
- FR3: The system shall capture sensor outputs including RGB images, depth maps and segmentation data.
- FR4: The system shall apply SHA-256 cryptographic hashing to each generated dataset batch to verify data integrity.
- FR5: The system shall encrypt generated dataset batches using AES-256 before storage or inter-component transfer.
- FR6: The system shall enforce RBAC on all simulation outputs, restricting access based on predefined user roles.
- FR7: The system shall apply privacy-preserving mechanisms during the synthetic data generation process.

- FR8: The system shall store the generated datasets in a structured format suitable for machine learning model training.
- FR9: The system shall support the export of generated datasets for analysis and machine learning experimentation.

5.3 Non-Functional Requirements

Security

Cybersecurity controls (integrity verification, encryption and RBAC) shall be implemented throughout the pipeline to protect against unauthorized access, data tampering and information leakage, consistent with security recommendations in recent AV security research [17], [21].

Performance

The system shall generate large volumes of synthetic datasets efficiently, with security control overhead (hashing and encryption) not significantly delaying the pipeline.

Scalability

The architecture shall support large-scale dataset generation across multiple simulation scenarios while maintaining security controls on all batches.

Usability

The system shall provide a user-friendly interface with clear role-based access management for authorized users.

Reliability

The system shall consistently produce accurate, secure and usable datasets for machine learning training purposes.

6. Commercial Plan

6.1 Target Market and Customer Profile

The proposed framework has applicability across industries involved in autonomous vehicle development and AI research. The primary target market comprises autonomous vehicle manufacturers, AI research laboratories and technology companies developing self-driving systems. The typical customer profile includes AI researchers, data scientists, cybersecurity engineers and AV engineers requiring large volumes of secure, labeled datasets.

6.2 Value Proposition

The proposed system offers a privacy-aware and cybersecurity-hardened synthetic data generation solution. Unlike conventional approaches where security is treated separately, this system embeds cryptographic integrity verification, encryption and access control directly into the generation process — addressing growing regulatory and security concerns highlighted in recent autonomous driving data security research [18].

6.3 Revenue Model

The solution may be commercialized as a software platform or dataset generation service. Organizations could subscribe to generate customized synthetic datasets. Additional revenue may come from dataset generation services tailored to specific driving scenarios or machine learning requirements, or through integration with existing simulation platforms as a security extension.

6.4 Cost Estimation

Development costs primarily involve software development, simulation infrastructure and computational resources. Since CARLA and cryptographic libraries are open-source, the initial development cost remains relatively low. Main expenses include cloud computing for largescale simulations, dataset storage and system maintenance.

6.5 Pricing Strategy

A subscription-based pricing model may be adopted, with fees varying based on the number of simulation scenarios, dataset size and computational resources required. A pay-per-dataset model may also be implemented for organizations with specific, one-time requirements.

6.6 Competitive Advantage

The proposed system offers competitive advantage by combining synthetic data generation with integrated privacy protection and cybersecurity mechanisms — a combination that is largely absent from current simulation platforms. Most existing tools treat security as a postgeneration concern, whereas this framework embeds security controls at the point of generation, enabling safer dataset sharing and compliance with emerging data security regulations [18], [21].

6.7 Intellectual Property Considerations

The framework may incorporate innovative techniques for integrating cybersecurity and privacy-preserving mechanisms within simulation-based dataset generation. These may be

safeguarded through software copyrights, patents or proprietary algorithms to retain commercial value and support future research collaborations.

7. Budget and Justifications

The estimated budget for completing the project is approximately LKR 48,000. The budget allocation is presented in Table 1.

Table 1: The Budget

Item	Estimated Cost (LKR)	Justification
Cloud GPU Compute (AWS, approx. 200 hours)	LKR 35,000	DP training and multiple ablation experiments across different epsilon settings require substantial GPU time before the final generation run.
Cloud Storage (500 GB)	LKR 5,000	Storage for synthetic dataset batches, model checkpoints, encrypted dataset files, hash manifest logs and privacy budget logs across all experimental runs.
Opacus and TensorFlow Privacy (open source)	LKR 0	Both libraries are freely available under open-source licenses and do not require any paid subscription.
Python cryptographic libraries (hashlib, PyCryptodome)	LKR 0	Open-source libraries used to implement SHA-256 hashing and AES-256 encryption within the secure data generation pipeline. No cost incurred.
Research paper access (IEEE Xplore, ACM DL)	LKR 8,000	Key papers on differentially private GANs, AV data privacy and secure pipeline design sit behind paywalls and require institutional or individual purchase.
Human effort (research and implementation)	LKR 0	All research and implementation work is covered within the academic project scope.
Total	**LKR 48,000**	

8. Work Breakdown Structure

The Work Breakdown Structure (WBS) outlines the distribution of project tasks among group members to ensure balanced workload and efficient project execution.

Table 2: Work Breakdown Structure of Individuals

Member	Role	Phase	Weeks	Key Tasks
IT22309556	Privacy Integration	1	1–3	Literature review on DP mechanisms. Library selection (Opacus vs TF Privacy). Define epsilon range for ablation.
IT22309556	Privacy Integration	2	4–7	Build DP training wrapper. Implement Privacy-Aware Calibration Module with Laplace noise. Run ablation experiments.
IT22309556	Privacy Integration	3	8–10	Full-scale generation run under epsilon-DP. Monitor epsilon consumption. Confirm budget not exceeded.
IT22309556	Privacy Integration	4	11–14	Coordinate with IT22066916 on MIA tests.
IT22541284	Secure Dataset Generation / Simulation	1	1–3	Set up CARLA simulator environment. Define scene and sensor configurations. Design RBAC policy for simulation outputs.
IT22541284	Secure Dataset Generation / Simulation	2	4–8	Run CARLA simulation using privatized calibration statistics. Implement SHA-256 integrity hashing on each dataset batch. Implement AES-256 encryption of generated batches. Enforce RBAC credential checks within the CARLA API wrapper.
IT22541284	Secure Dataset Generation / Simulation	3	9–12	Validate dataset format, sensor coverage and hash manifests. Audit RBAC logs. Deliver final encrypted and integrity-verified synthetic dataset to IT22110220.
IT22541284	Secure Dataset Generation / Simulation	4	12–14	Document cybersecurity layer design and implementation. Contribute security section to group thesis.

IT22110220	Utility Evaluation	1	1–4	Define utility metrics (mAP, FID). Set up AV model training pipeline and baseline benchmarks.
IT22110220	Utility Evaluation	2	8–11	Train AV model on synthetic dataset. Measure mAP and FID against non-private baseline.
IT22110220	Utility Evaluation	3	12–14	Report utility trade-off results. Contribute utility analysis to group thesis.
IT22066916	Risk and Compliance	1	1–3	Review GDPR, CCPA, and Sri Lanka PDPA requirements. Define compliance checklist.
IT22066916	Risk and Compliance	2	9–12	Run membership inference and reidentification attack tests on DP synthetic data from IT22309556.
IT22066916	Risk and Compliance	3	12–14	Compile attack results and compliance report. Contribute risk and regulatory sections to group thesis.

Table 3: Timeline Alignment

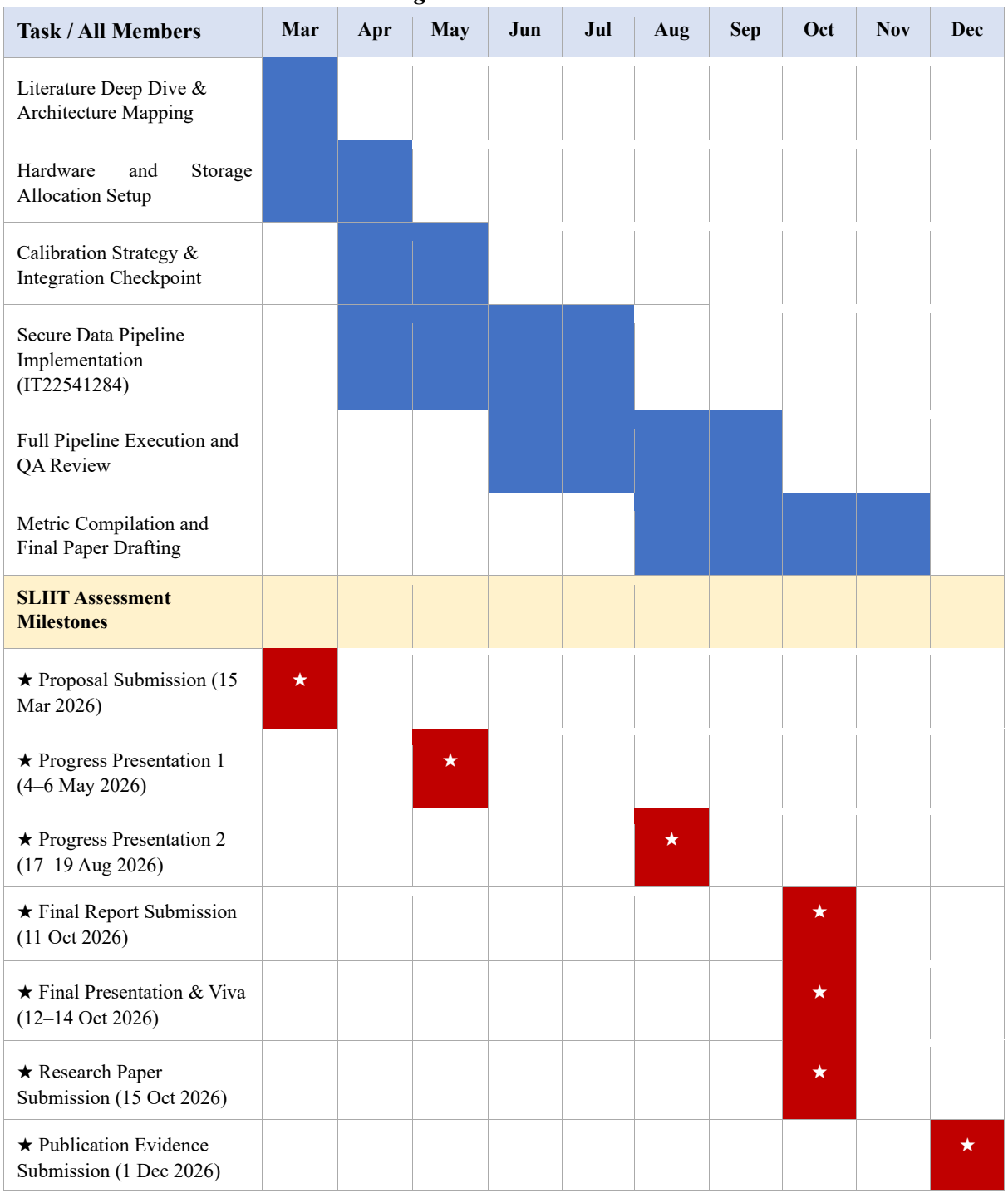
Member	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
IT22309556	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IT22541284	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IT22110220	1	2	3	4				8	9	10	11	12	13	14
IT22066916	1	2	3						9	10	11	12	13	14

Table 4: Balanced Workload Distribution

Member	Active Weeks	Phases	Primary Deliverables
IT22309556	1–14 (14 weeks)	4	epsilon-DP training wrapper, calibration module, ablation results, privacy budget log
IT22541284	1–14 (14 weeks)	4	CARLA simulation setup, secure synthetic dataset batches (SHA-256 hashed, AES-256 encrypted), RBAC audit logs, validated dataset delivery
IT22110220	1–4, 8–14 (11 weeks)	4	Utility metric framework, AV model training results, mAP and FID analysis report
IT22066916	1–3, 9–14 (9 weeks)	4	Compliance checklist, MIA and re-identification attack results, regulatory compliance report

9. Gantt Chart

Figure 3: Gantt Chart



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